Mobile Lidar Accuracy Testing SOP

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1. Purpose and method

The most appropriate method is an independent checkpoint test: survey at least 30 checkpoints to a higher accuracy than the lidar, compare each one to the adjusted point cloud, and report the result as RMSE. This follows the ASPRS Positional Accuracy Standards for Digital Geospatial Data, Edition 2, Version 2 (2024).

The method has four rules that separate it from the FDOT test:

- **Count:** at least 30 checkpoints for vertical and 30 for horizontal. FDOT asks for 20.
- **Independence:** checkpoints are never used to adjust, register or calibrate the point cloud.
- **Checkpoint quality:** checkpoints are surveyed at least twice as accurately as the accuracy being tested.
- **Statistic:** results are stated as RMSE, with the checkpoint survey error added in. No 95% confidence figure is used.

This SOP applies to vehicle, hi-rail and backpack mobile lidar on roads, corridors and sites. It covers testing your own data and auditing another firm's data. Sources: ASPRS Edition 2 overview and summary of Edition 2 changes. Confirm section numbers and exact wording against the standard itself before contract use.

2. Accuracy targets

Set the target RMSE before fieldwork, because it fixes how accurate the checkpoints must be. FDOT limits are stated at 95% confidence, so they are converted here by dividing by 1.96 (vertical) and 1.7308 (horizontal).

FDOT type	FDOT limit at 95%	Vertical RMSE target	Horizontal RMSE target	Checkpoint RMSE needed (vertical / horizontal)
Type A	0.06 ft	0.031 ft (0.93 cm)	0.035 ft (1.06 cm)	0.47 cm / 0.53 cm
Type B	0.10 ft	0.051 ft (1.56 cm)	0.058 ft (1.76 cm)	0.78 cm / 0.88 cm

FDOT type	FDOT limit at 95%	Vertical RMSE target	Horizontal RMSE target	Checkpoint RMSE needed (vertical / horizontal)
Type C	Per project scope	Per scope	Per scope	Half the scoped RMSE

- The conversion assumes errors are normally distributed with no bias. It is an equivalence for planning, not an ASPRS rule.
- ASPRS accuracy classes are named by their RMSE in centimeters. A Type A survey corresponds to roughly a 1 cm class and Type B to roughly a 1.5 to 2 cm class.
- Type A checkpoints need better than 5 mm accuracy. RTK GNSS cannot deliver that; use digital leveling and total station work tied to project control.

3. Checkpoints

Plan the checkpoints before the lidar is collected, and keep their coordinates away from the processing team until the point cloud is final.

Number

- At least 30 vertical checkpoints on hard, non-vegetated surfaces. ASPRS scales the count up to 120 for areas above 1,000 km², which no road corridor reaches.
- At least 30 horizontal checkpoints. A point can serve both tests if it is a well-defined target.
- Add 10% spares so a blocked or disturbed point does not drop the count below 30.
- Vegetated surfaces, if tested, need their own 30 points. The result is reported but is not a pass/fail criterion.

Placement

- Spread points along the full length and both sides of the corridor. Where FDOT spacing applies, 1,000 ft or less for Type A already yields 30 points on about 5.5 miles.
- On short projects, tighten the spacing until 30 points fit. Do not cluster them at one end.
- Include the weak areas: bridges, underpasses, tree canopy, tall buildings and the far ends from the base station.
- Vertical points go on flat, smooth pavement or concrete, away from curbs, joints, grates and edges.

- Horizontal points must be identifiable in the cloud: painted chevrons or checkerboard targets, or sharp paint-stripe corners visible in intensity.

Survey method

- Survey to at least twice the accuracy of the target in section 2.
- Tie to the same project control and datum as the lidar: horizontal datum, vertical datum, geoid model, epoch and units.
- Measure each point twice, independently, and record the difference.
- State the checkpoint RMSE in the surveyor's report. It is an input to the final computation.

Independence

- A point used for trajectory adjustment, registration or transformation is a control point, not a checkpoint.
- If a checkpoint is later used to fix the data, replace it with a new one before retesting.

4. Point cloud preparation

Test only the final, adjusted point cloud that will be delivered. Complete these steps first and record each in the processing log.

1. Process the trajectory with base station data. Record satellite geometry, baseline lengths and any GNSS outages, with their locations.
2. Apply the boresight and lever-arm calibration. Keep the before and after calibration records.
3. Check pass-to-pass agreement where drive lines overlap. Resolve mismatches before adjusting to control.
4. Adjust the cloud to the ground control targets. Record the residual at every control point.
5. Confirm datum, geoid model, projection and units match the checkpoint survey.
6. Classify the cloud, at minimum separating ground and hard surface from noise, vehicles and vegetation.
7. Remove noise and moving-object returns around each checkpoint location.
8. Freeze the cloud. Record the file names, date and software versions. Any later change means a full retest.

Control residuals from step 4 are not an accuracy result. They show how well the cloud was fitted, and they always look better than an independent test.

5. Test procedure

Run the vertical, horizontal and pass-to-pass tests separately, and keep every checkpoint in the table unless there is a documented blunder.

Vertical test

1. For each checkpoint, select the hard-surface points within a small horizontal radius, typically 0.25 to 0.5 ft.
2. Fit a plane to those points, or build a triangulated surface from them, and take the elevation at the checkpoint's horizontal position.
3. Do not use the single nearest point. One noisy return would stand in for the surface.
4. Require a minimum number of points in the fit, for example 10. Flag any checkpoint that falls short.
5. Compute the error as lidar elevation minus surveyed elevation.

Horizontal test

1. Find each target or stripe corner in the cloud using intensity.
2. Extract its position by fitting the target shape or intersecting the fitted edges. Do not pick a single point by eye.
3. Compute the errors as lidar minus surveyed, separately for easting and northing.
4. If fewer than 30 targets can be identified, the horizontal result must be reported with the reduced-count statement in section 7.

Pass-to-pass test (relative accuracy)

1. In overlap areas, compare surfaces from separate passes on smooth pavement.
2. Sample at least 30 locations spread along the project.
3. Report the RMS and maximum difference. A large difference signals trajectory or calibration problems even when the checkpoint test passes.

Blunders

- Remove a checkpoint only for a proven cause: disturbed mark, misidentified target, obstruction in the cloud, or a survey error shown by the repeat measurement.
- Document each removal with the reason and the value removed. Never remove a point only because its error is large.

- If removals drop the count below 30, survey replacements.

6. Computation

Compute the fit to checkpoints first, then add the checkpoint survey error to get the product accuracy. The product accuracy is the number that is compared to the target.

Fit to checkpoints, for n checkpoints with errors e:

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RMSE_{V1} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_{z,i}^{2}} \qquad RMSE_{H1} = \sqrt{RMSE_{x}^{2} + RMSE_{y}^{2}}
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Product accuracy, where the second term is the checkpoint survey RMSE from the surveyor's report:

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RMSE_{V} = \sqrt{RMSE_{V1}^{2} + RMSE_{V2}^{2}} \qquad RMSE_{H} = \sqrt{RMSE_{H1}^{2} + RMSE_{H2}^{2}}
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Three-dimensional accuracy:

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RMSE_{3D} = \sqrt{RMSE_{H}^{2} + RMSE_{V}^{2}}
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Worked check, in centimeters:

Quantity	Input	Result
Vertical fit to 32 checkpoints	measured	0.80
Vertical checkpoint survey RMSE	from surveyor	0.40
Vertical product RMSE	$\sqrt{(0.80^2 + 0.40^2)}$	0.89
Horizontal fit	$\sqrt{(0.60^2 + 0.70^2)}$	0.92
Horizontal checkpoint survey RMSE	from surveyor	0.50
Horizontal product RMSE	$\sqrt{(0.92^2 + 0.50^2)}$	1.05
3D RMSE	$\sqrt{(1.05^2 + 0.89^2)}$	1.38

In this example the vertical result of 0.89 cm meets the Type A target of 0.93 cm, and the horizontal result of 1.05 cm meets 1.06 cm. Without the checkpoint term the margins would look larger than they are.

Also compute and record the mean, median, standard deviation, minimum and maximum error for each test. A mean error larger than about a quarter of the RMSE suggests a bias, such as a datum, geoid or antenna-height problem. Find the cause; do not subtract it out.

7. Acceptance and reporting

The data passes when the product RMSE is at or below the target RMSE, with at least 30 checkpoints in that test.

Outcome	Action
RMSE at or below target, 30 or more checkpoints	Accept. Issue the standard "tested to meet" statement
RMSE at or below target, fewer than 30 checkpoints	Report with the reduced-count statement, naming the number used. This is not full compliance
RMSE above target, bias present	Investigate datum, geoid, control and calibration. Correct, then retest with fresh checkpoints
RMSE above target, no bias	Reprocess the trajectory, add control or recollect the weak sections. Retest
Pass-to-pass difference large, checkpoint test passes	Investigate before delivery. Report the finding either way

The accuracy report must contain:

- Project name, limits, dates of lidar collection and checkpoint survey.
- System, software versions and the frozen file list.
- Datum, geoid model, projection and units.
- Checkpoint survey method and its RMSE.
- A table of every checkpoint: ID, surface type, surveyed and lidar coordinates, and errors.
- Removed checkpoints, with reasons.
- Statistics from section 6 and the pass-to-pass results.

- The ASPRS reporting statement. It names the standard and edition, the accuracy class tested, and the tested RMSE. Copy the exact wording from the reporting section of the standard.
- If FDOT also applies, the 95% values (1.96 × vertical RMSE; 1.7308 × horizontal RMSE) and the Type achieved.
- Signature and seal of the Florida Professional Surveyor and Mapper in responsible charge.

8. Auditing another firm's survey

A firm's compliance can be judged from its sealed survey report and checkpoint table, without the point cloud, as long as the table lists each checkpoint error.

1. Obtain the sealed survey report, checkpoint table, control report and processing log. Request templates are in the Records requests tab.
2. Count the independent checkpoints. Exclude any point that also appears in the control or transformation list.
3. Recompute the fit RMSE from the listed errors. Do not rely on the firm's summary figure.
4. Find the checkpoint survey RMSE in the report. If none is stated, record that as a deficiency and estimate it from the survey method.
5. Compute the product RMSE with the section 6 formulas and compare it to the section 2 targets.
6. Check the mean error for bias and the checkpoint map for clustering or untested weak areas.
7. Assign a finding from the table below.

Finding	Condition
Meets ASPRS and FDOT	30 or more independent checkpoints; product RMSE at or below target; 95% value within the FDOT limit
Meets FDOT only	20 to 29 checkpoints, 95% value within the FDOT limit. ASPRS result reported with the reduced-count statement
Does not meet	Product RMSE or 95% value above the limit

Finding	Condition
Cannot be determined	No per-point errors, checkpoints not independent, or no report

If the point cloud is also obtained, an independent retest is possible by surveying new checkpoints. Use only surfaces unchanged since the lidar was collected. Resurfaced pavement and rebuilt sections must be excluded.